

TEAM MEMBER 1

Mathematical Modeling, Numerical Analysis, and Theory

Task Assignment | Mathematical Analysis and Experiment Specification

Prerequisite: The team has already prepared a high-accuracy reference solution. This task does not repeat the construction of the reference solution. All subsequent error, stability, and efficiency experiments must use the reference trajectory confirmed by the team.

Your responsibility: Explain the “why”: establish the mathematical model and numerical analysis framework for the Robertson problem; derive the numerical methods and the theory of stability, stiffness, convergence, and Newton's method; and specify the numerical experiment corresponding to each theoretical conclusion. Do not take responsibility for the main code implementation.

1. Robertson Model and Basic Properties

Starting from the system provided in the project, present the model, initial conditions, and physical interpretation:

$$\begin{aligned}y_1' &= -0.04 y_1 + 10^4 y_2 y_3 \\y_2' &= 0.04 y_1 - 10^4 y_2 y_3 - 3 \times 10^7 y_2^2 \\y_3' &= 3 \times 10^7 y_2^2, y(0) = (1, 0, 0)^T\end{aligned}$$

- Derive $y_1' + y_2' + y_3' = 0$, and hence the conservation law $y_1 + y_2 + y_3 = 1$.
- Explain that y_1 , y_2 , and y_3 represent concentrations and should therefore satisfy componentwise non-negativity: $y_i(t) \geq 0$.
- State that conservation and positivity will subsequently be used as separate diagnostics.

2. Three Numerical Methods: Formulas and Theoretical Orders

Present the three basic methods required by the Project Brief and relate them to the Robertson system.

Explicit Euler

$$y_{n+1} = y_n + h f(t_n, y_n), \text{ global error} = O(h)$$

Classical RK4

$$\begin{aligned}k_1 &= f(t_n, y_n) \\k_2 &= f(t_n + h/2, y_n + h k_1/2) \\k_3 &= f(t_n + h/2, y_n + h k_2/2) \\k_4 &= f(t_n + h, y_n + h k_3) \\y_{n+1} &= y_n + h(k_1 + 2k_2 + 2k_3 + k_4)/6, \text{ global error} = O(h^4)\end{aligned}$$

Implicit Euler

$$y_{n+1} = y_n + h f(t_{n+1}, y_{n+1}), \text{ global error} = O(h)$$

3. Newton Iteration for Implicit Euler

$$F(z) = z - y_n - h f(t_{n+1}, z) = 0$$

$$J_F(z) = I - h J_f(t_{n+1}, z)$$

$$J_F(z^{(k)}) \delta^{(k)} = -F(z^{(k)})$$

$$z^{(k+1)} = z^{(k)} + \delta^{(k)}$$

- Explain that n is the time-step index and k is the Newton iteration index.
- Explain why the code should solve the linear system $J_F \delta = -F$ rather than explicitly form the inverse of J_F .
- Explain that an overly loose Newton tolerance introduces nonlinear-solve error, while an overly tight tolerance increases cost; solving to machine precision is generally unnecessary.

4. Convergence Theory and Experimental Criteria

$$p_{\text{obs}} = \log(E_h / E_{h/2}) / \log(2)$$

- Use at least four successively halved step sizes: h , $h/2$, $h/4$, and $h/8$.
- Theoretical expectations: $p \approx 1$ for Explicit Euler (EE) and Implicit Euler (IE), and $p \approx 4$ for RK4.
- Explain why the expected error reduction by a factor of 2 or 16 becomes clear only after entering the asymptotic regime.
- Provide the implementation team with an explicit error definition and require error-versus-step-size log-log plots for at least two methods.

5. Absolute Stability of the Three Methods

Explicit Euler: $R(z) = 1 + z$; stable if $|1 + z| \leq 1$

RK4: $R(z) = 1 + z + z^2/2 + z^3/6 + z^4/24$; stable if $|R(z)| \leq 1$

Implicit Euler: $R(z) = 1/(1 - z)$; stable if $|1/(1 - z)| \leq 1$

- Derive the three stability functions rather than merely stating the formulas.
- Explain why the A-stability of IE is important for the stiff Robertson system.
- Explicitly require the implementation team to plot the absolute-stability regions of all three methods.

6. Robertson Jacobian

$$J_f(y) = \begin{bmatrix} -0.04 & 10^4 y_3 & 10^4 y_2 \\ 0.04 & -10^4 y_3 - 6 \times 10^7 y_2 & -10^4 y_2 \\ 0 & 6 \times 10^7 y_2 & 0 \end{bmatrix}$$

- Explain how each partial derivative is obtained, and verify the signs and the $6 \times 10^7 y_2$ terms.

7. Structural Zero Eigenvalue and Initial Degeneracy

- Use the conservation law to show that the full 3×3 Jacobian always has a structural zero eigenvalue; emphasize that this is not a numerical error.
- Verify that the eigenvalues at $y(0) = (1, 0, 0)^T$ are $\{-0.04, 0, 0\}$.

- Explain that there is only one nonzero mode at $t = 0$, so the stiffness ratio is uninformative at the initial point. The plot should begin at the first positive logarithmically spaced time.

8. Stiffness Ratio and Reduced Formulation

$$S(t) = \max_{\lambda_j \neq 0} |\operatorname{Re} \lambda_j(t)| / \min_{\lambda_j \neq 0} |\operatorname{Re} \lambda_j(t)|$$

- Explain the fast and slow modes, and how $S \gg 1$ relates to stiffness arising from multiple time scales.
- Define an explicit numerical threshold for identifying zero eigenvalues, to be used consistently throughout the code.
- Optionally provide a reduced formulation using $y_3 = 1 - y_1 - y_2$, and use the two eigenvalues of the 2×2 reduced Jacobian for an equivalent analysis.

9. Robertson-Specific Stability Restriction for Explicit Euler

For negative real λ : $|1 + h\lambda| \leq 1 \Rightarrow h < 2/|\lambda|$
 Using $|\lambda|_{\max} \approx 3.39 \times 10^3$ gives $h_{\text{pred}} \approx 5.9 \times 10^{-4}$

- Emphasize that 5.9×10^{-4} is only a local frozen-Jacobian estimate, not a nonlinear stability theorem.
- Explicitly require the implementation team to perform an empirical step-size sweep near this value and check for three phenomena: instability, inaccuracy, and negative concentrations.

10. Consistent Mathematical Definitions of Diagnostics

$$\begin{aligned} E_{\text{full}} &= \max_i \|y_{\text{num}}(t_i) - y_{\text{ref}}(t_i)\|_2 \\ E_{40} &= \|y_{\text{num}}(40) - y_{\text{ref}}(40)\|_2 \\ E_{\text{cons}} &= \max_i |y_1(t_i) + y_2(t_i) + y_3(t_i) - 1| \\ y_{\min} &= \min_{i,j} y_j(t_i) \end{aligned}$$

- Provide the project's criteria for classifying results as stable, non-negative, and accurate, so that the code and report do not use different standards.
- Emphasize that conservation, positivity, and accuracy must be assessed separately.

11. Matched-Error Comparison

- Explain why using the same h does not provide a fair efficiency comparison. EE should be compared with IE/RK4 at matched accuracy, with $E_{\text{method1}} \approx E_{\text{method2}}$.
- Explain that “did not blow up” demonstrates only boundedness/stability, not accuracy.
- Specify the cost measure for the implementation team - RHS evaluations are recommended, with Newton iterations reported additionally for IE - and identify the costs excluded from the measure.

12. Newton Tolerance Sweep

- Fix a reasonable IE step size. Suggested tolerances are $\text{tol} = 10^{-4}, 10^{-6}, 10^{-8}, 10^{-10}$, and 10^{-12} .
- Explain theoretically when the nonlinear-solve error has become much smaller than the time-discretization error.
- Focus on explaining why $y_2(40)$ is a sensitive diagnostic and why the result should stabilize as the tolerance is tightened.

13. Mathematical Scheme for Adaptive Step-Size Control

- Provide a basic adaptive scheme; RK4 with step doubling is recommended.

Compare one step of size h with two successive steps of size $h/2$ to estimate local error.

- Explain how to update h_{new} using the error estimate and tolerance, and define the acceptance and rejection rules.
- Require the implementation team to demonstrate that $h(t)$ actually varies and to use the reference solution to verify the relationship between the actual error and the target tolerance.

14. Required Numerical Experiments

- **Convergence:** Use at least four successively halved step sizes; record E_{full} and the observed order.
- **Stability regions:** Provide three plots for EE, RK4, and IE, or a clearly combined plot.
- **Eigenanalysis:** Calculate the two nonzero eigenvalues and $S(t)$ along the reference trajectory.
- **EE failure sweep:** Check stability, accuracy, and non-negativity around $h \approx 5.9 \times 10^{-4}$.
- **IE matched-error comparison:** Compare IE with EE at similar E_{full} values.
- **Newton tolerance sweep:** Record $y_2(40)$, E_{full} , E_{cons} , and Newton iterations.
- **Adaptive method:** Show $h(t)$, accepted and rejected steps, and error relative to the reference solution.
- **Cost versus accuracy:** Compare RHS evaluations and Newton iterations using a common accuracy target.

15. Final Deliverables and Acceptance Criteria

Deliverable	Required Content	Acceptance Criteria
Mathematical derivation document	Model; three methods; Newton's method; convergence; stability; Jacobian; eigenvalues; stiffness; adaptive methods; and cost framework.	Formulas are correct; every conclusion is connected to the Robertson problem.
Experiment specification checklist	The data, plots, tables, and assessment criteria required for each theoretical conclusion.	The implementation team can implement the specifications directly without guessing definitions.
Results interpretation draft	An explanation of theory versus experiment once the numerical results are available.	Explains the reasons for agreement or disagreement rather than merely restating the plots.
Q&A preparation	Ability to explain the model, formulas, stiffness, Newton's method, stability, and convergence.	The team member can independently answer cross-questions during the seminar.

Core principle: Organize every part as “Theory → Prediction → Required numerical experiment → Interpretation.” Do not merely reproduce textbook formulas.